Stat 266 – Unit 8 – Video Set 3 – Class Examples

Summary:

Recall: • For a random variable Y, the k-th **raw moment** is $\mu_k' = E\left[Y^k\right]$.

• The k-th **sample raw moment**, based on a random sample of size n is $m_k' = \frac{1}{n}\sum_{i=1}^{n} Y_i^k$.

$m_1' = \frac{1}{n}\sum_{i=1}^{n} Y_i^1 = \bar{Y}$

$m_2' = \frac{1}{n}\sum_{i=1}^{n} Y_i^2$

Method of Moments Estimation:

Choose as estimates for $\theta = (\theta_1, \theta_2, \ldots, \theta_t)$, the values that are the solutions to

$\mu_k' = m_k', \quad k = 1, 2, \ldots$.

Let $\hat{\theta}_{MM}$ denote the Method of Moments Estimator for θ.

Quick Examples:

- $Y \sim Exp(\mu_Y = \beta) \Rightarrow \mu_Y = \beta \Rightarrow m_1' = \bar{Y}$

  $\Rightarrow \hat{\beta}_{MM} = \bar{Y}$

- $Y \sim Uniform(0, \theta) \Rightarrow \mu_Y = \frac{\theta}{2} \Rightarrow m_1' = \bar{Y}$

  $\Rightarrow \hat{\theta}_{MM} = 2\bar{Y}$

- $Y \sim Beta(\alpha, \beta = 2) \Rightarrow \mu_Y = \frac{\alpha}{\alpha+2} \Rightarrow \bar{Y} \Rightarrow \hat{\alpha}_{MM} = \frac{2\bar{Y}}{1-\bar{Y}}$.

In general

$$m_k' = \frac{1}{n} \sum_{i=1}^{n} Y_i^k .$$

$E[Y^k] \Rightarrow$ population average of Y^k.

$$E[m_k'] = E\left[\frac{1}{n} \sum_{i=1}^{n} Y_i^k\right] = \frac{1}{n} n E[Y^k]$$

$$= E[Y^k].$$

e.g. Regardless of the distribution of Y,

$$\hat{\mu}_{mm} = \bar{Y}.$$

Example A:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}\dfrac{\alpha\beta^{\alpha}}{y^{\alpha+1}}, & 0<\beta<y,\alpha>0\\ 0 & \text{, else}\end{cases}.$$

$$\mu_1' = E[Y] = \int_{\beta}^{\infty} \frac{\alpha\beta^{\alpha}}{y^{\alpha}}\,dy = \frac{\alpha\beta}{\alpha-1}$$

a) Suppose α is a known constant. Find $\hat{\beta}_{MM}$.

b) Suppose β is a known constant. Find $\hat{\alpha}_{MM}$.

a) $\dfrac{\alpha\beta}{\alpha-1} \Rightarrow \bar{Y} \Rightarrow \hat{\beta}_{MM} = \dfrac{\alpha-1}{\alpha}\bar{Y}$ } Only an estimator if α is known.

b) $\dfrac{\alpha\beta}{\alpha-1} \Rightarrow \bar{Y} \Rightarrow \hat{\alpha}_{MM} = \dfrac{\bar{Y}}{\bar{Y}-\beta}$

c) 1) $\mu_Y = \mu_1' = \dfrac{\alpha\beta}{\alpha-1} \Rightarrow m_1' = \bar{Y}$, $\boxed{\dfrac{\alpha\beta}{\alpha-1} = \bar{Y}}$

2) $E[Y^2] = \mu_2' = \dfrac{\alpha\beta^2}{\alpha-2} \Rightarrow m_2' = \dfrac{1}{n}\sum_{i=1}^{n} Y_i^2$, $\boxed{\dfrac{\alpha\beta^2}{\alpha-2} = m_2'}$

Two equations, two unknowns. Solve for α and β as functions of $\bar{Y}$ and m_2'.

$$\Rightarrow \hat{\alpha}_{MM} = 1 + \sqrt{1+\frac{n}{n-1}S^2}, \quad S^2 = \frac{1}{n-1}\sum(Y_i-\bar{Y})^2 = \frac{n}{n-1}\left(m_2' - \bar{Y}^2\right).$$

$$\hat{\beta}_{MM} = \frac{\bar{Y}\sqrt{1+\frac{n}{n-1}S^2}}{1+\sqrt{1+\frac{n}{n+1}S^2}}$$

Example B:

Suppose $Y_1, \ldots, Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y) = \begin{cases} (1-\beta)y^{-\beta} & , 0 < y < 1, \ \beta < 1 \\ 0 & , \text{else} \end{cases}.$$

Find the method of moments estimator of β.

$$\mu_1' = E[Y] = \int_0^1 (1-\beta) y^{-\beta+1} \, dy = \frac{1-\beta}{2-\beta} \Rightarrow \bar{Y}$$

$$\hat{\beta}_{MM} = \frac{1-2\bar{Y}}{1-\bar{Y}}.$$

Example C:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}\alpha\left(y^2-\frac{1}{3}\right)+\frac{1}{2} & , -1<y<1,\ -\frac{3}{4}<\alpha<\frac{3}{2}\\ 0 & , else\end{cases}$$

Find the method of moments estimator for α.

$$\mu_1' = E[Y] = \int_{-1}^{1} \alpha y^3 - \frac{\alpha}{3}y + \frac{1}{2}y \, dy = 0 \quad \text{(not helpful)}$$

$$\mu_2' = E[Y^2] = \int_{-1}^{1} \alpha y^4 - \frac{\alpha}{3}y^2 + \frac{1}{2}y^2 dy$$

$$= \frac{8\alpha}{45} + \frac{1}{3} \quad \Rightarrow M_2' = \frac{1}{n}\sum_{i=1}^{n} Y_i^2$$

$$\Rightarrow \hat{\alpha}_{MM} = \frac{45}{8}M_2' - \frac{15}{8} = \frac{45}{8n}\sum_{i=1}^{n} Y_i^2 - \frac{15}{8}.$$

Example D:

Suppose $Y_1, \ldots, Y_n$ is a random sample of independent and identically distributed $Geometric(p)$ random variables. Find an unbiased estimator for p as a function of $\hat{p}_{MM}$.

$$\mu_1' = E[Y] = \frac{1}{p} \Rightarrow \bar{Y} \qquad \Rightarrow \hat{p}_{MM} = \frac{1}{\bar{Y}}.$$

Is $\hat{p}_{MM}$ unbiased?

$$E[\hat{p}_{MM}] = E\left[\frac{1}{\bar{Y}}\right] = E\left[\frac{n}{\sum Y_i}\right] = n\,E\left[\frac{1}{\sum Y_i}\right] = ?$$

<u>Note:</u> $\sum_{i=1}^{n} Y_i \sim NegBin(r = n, p). \Rightarrow X$

$$\Rightarrow E\left[\frac{1}{X}\right] = \sum_{x=n}^{\infty} \frac{1}{x}\binom{x-1}{n-1} p^n (1-p)^{x-n} = ?$$

Re-visit later.

Example E:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}\frac{1}{\beta}e^{-\frac{(y+\beta)}{\beta}} & , y>-\beta \\ 0 & , else\end{cases} \quad , \quad \beta > 0$$

Find the method of moments estimator for β.

$$\mu_1' = E[Y] = \int_{-\beta}^{\infty} \frac{y}{\beta} e^{-\frac{1}{\beta}(y+\beta)}\, dy$$

Let $U = Y+\beta$. $y = h^{-1}(u) = u-\beta$, $\frac{dy}{du} = 1$.

$\Rightarrow f_U(u) = f_Y(u-\beta)\cdot 1 = \frac{1}{\beta} e^{-u/\beta}$, $u>0$.

$U \sim Exp(\beta)$.

- $Y = U-\beta$, $E[Y] = \mu_U - \beta = 0$. (not helpful)
- $E[Y^2] = E[(U-\beta)^2] = E[U^2] - 2\beta E[U] + \beta^2$

$= 2\beta^2 - 2\beta^2 + \beta^2 = \beta^2$. $\Rightarrow \mu_2'$

$\therefore \hat{\beta}_{MM} = \sqrt{\mu_2'} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} Y_i^2}$.